Presenter Notes

Functional Forecasting of Turkish Electricity Load Curves — Zeynep Bozoklu

Slide 1 — Title

Introduce the thesis topic: applying Functional Data Analysis to forecast Turkey's daily electricity load curve. Core result: our model beats the official TEİAŞ forecast by 19% with near-zero p-value. Walk through 14 slides in roughly 20 minutes.

Slide 2 — The Forecasting Object

Point to the overnight trough (around hour 4–5) and the broad daytime plateau (hours 9–20). The shape is structurally the same every day; what changes is the level and sharpness. Note how the official forecast (dashed) tracks the actual well on average but lies systematically below it — we will quantify this underprediction on the next slide.

Slide 3 — Why Accuracy Matters

The official forecast always undershoots actual demand. The gap widened sharply in 2024. Our rolling-origin model reduces the average bias by 78%. Just state the empirical fact — do not speculate about why the official underpredicts. This table motivates the research question: is there systematic improvement to be had?

Slide 4 — From 24 Numbers to a Curve: The FDA Pipeline

Walk through the boxes left to right, then down and back. The core idea: instead of 24 separate forecasting models, we compress to 2 scores, forecast those, and reconstruct the full curve. The B-spline step also lets us compute the derivative of the curve, capturing ramp speed — a feature we add later. This is the methodological contribution.

Slide 5 — Data

Data from EPIAS, Turkey's electricity market operator, via public API. Having both actual load and the official TEİAŞ day-ahead forecast is key: it gives us a directly comparable operational benchmark. No weather data is used in the main model — a deliberate choice we revisit when discussing ramp-hour losses.

Slide 6 — FPCA: Two Numbers Capture the Curve

Top panel (FPC1): nearly flat at 0.13–0.28, meaning a high score uniformly lifts the whole curve — the "how much electricity today" factor. Bottom panel (FPC2): positive at hours 0–5 and 18–23, negative daytime — the overnight-vs-daytime contrast. Three coloured lines (2019–21, 2020–22, 2022–24) nearly overlap, which empirically justifies using a fixed FPCA basis rather than re-estimating it dynamically.

Slide 7 — Score Forecasting Model

Lag 7 captures the dominant weekly pattern (Monday resembles last Monday); lag 1 adjusts for recent drift. All features in z_d are known before day d starts — no lookahead. Derivative features come from differentiating the B-spline curve, capturing how fast demand was changing yesterday. Key message to committee: holiday indicators are non-negotiable; the model is competitive only once they are included.

Slide 8 — Evaluation: Rolling-Origin Design

Stress the no-leakage principle: the model never sees the future. This mirrors real operational deployment. The Diebold-Mariano test is appropriate because we compare two forecasts for the same target over the same sample — it accounts for forecast error correlation. Citing Harvey, Leybourne and Newbold (1997) for the small-sample correction is good practice if the committee asks.

Slide 9 — Main Results

Walk the table top to bottom. The seasonal naïve shows the floor — the problem is hard. Pure VAR dynamics are worse than the official. The model crosses into positive territory only when holiday and load-shape features are added. Derivative features add another 2 points. Both gains are significant at near-zero p-value ($p \approx 10^{-135}$ and 10^{-163}). The main model (bold row) is the baseline contribution; the residual correction row previews the next extension.

Slide 10 — Where the Model Wins and Where It Loses

The ramp-hour dip at hours 6–9 is the key diagnostic. Demand nearly doubles during this window as factories start and HVAC switches on. We do not know what gives the official forecast its advantage at ramp hours — do not speculate. This gap motivates the residual correction step on the next slide.

Slide 11 — Residual Correction: Exploiting Persistent Errors

The ACF panel for Hour 6 is the punchline: base model errors are strongly autocorrelated (ACF ≈ 0.4 at lag 1, persisting 14 days). The orange series after correction drops sharply toward zero. The correction is estimated on past data only — no leakage. Key message: the residual step wins at the ramp hours (6–9) where the base model lost.

Slide 12 — Robustness Checks

Direct RF (forecasting each hour separately, no FPCA) beats FPCA-OLS at most daytime hours — this is a genuine finding. Do not downplay it. The FPCA-OLS advantage is at overnight hours. The right framing: both approaches beat the official; the comparative strengths differ by time of day.

Slide 13 — Thesis Road Map

The empirical core is complete. Open items: (1) a formal literature review situating this within FDA and electricity forecasting, (2) adding temperature data to address the ramp-hour weakness, and (3) probabilistic extensions if time allows. Be direct with the committee about what is done vs planned.

Slide 14 — References

Keep this slide up while answering questions. If asked about methodology, cite Kokoszka & Reimherr (2017). If asked about the DM test, cite Diebold & Mariano (1995). If asked about FDA applied to energy forecasting, Hyndman & Ullah (2007) is the closest precedent.